

ABDUL WAHAB

MECHANICAL DESIGN ENGINEER AND PRODUCT DESIGNER

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Portfolio: <https://wahababdull.github.io/CV/>

Research Interests: Design for autonomous manufacturing, mechanism design and automation systems, precision mechanical design and manufacturing process optimization, industrial product development for smart production environments, mechanical system design for reliability and maintainability.

Mechanical Design Engineer and Product Designer with 2.0 years of experience specializing in SolidWorks-based CAD modeling. Skilled in 3D part design, complex assemblies, and detailed manufacturing drawings in compliance with engineering standards. Proven experience designing industrial machinery, mechanical systems, and consumer products, with a strong focus on accuracy, manufacturability, and functional performance. Adept at transforming concepts into production-ready designs through creative problem-solving techniques and industry standards. I am ready to help turn ideas into reality.

SKILLS AND EXPERTISE

- CAD and Manufacturing Design
- Design Simulation and Manufacturability Analysis
- Manufacturing Process Design and Design for Manufacturability (DFM)
- Engineering Problem-Solving and Root Cause Analysis
- Process Data Analysis

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | Freelance (Upwork)

August 2025 – Present

- Delivered professional **mechanical and product design services** to international clients via **Upwork**.
- Created 3D CAD models, assemblies, 2D technical drawings, and photorealistic renders for prototyping and manufacturing.
- Designed components for consumer products and industrial applications, optimized for material selection and manufacturing processes.
- Communicated directly with clients to gather requirements, manage revisions, and meet project deadlines
- Maintained high client satisfaction through **quality delivery and clear technical communication**.
- **Skills:** SolidWorks, 3D CAD Modeling, 3D CAD Rendering, 2D Technical Drawing, Clear Communication, Detail Oriented Design, Prototyping Support

Mechanical Design Engineer | Self- Employed | D12 Markaz, Islamabad, Pakistan

December 2024 – Present

- Designed precise 3D mechanical parts, assemblies, and production drawings using SolidWorks.
- Completed multiple mini and industrial projects, including Centrifugal Pumps, Heat Exchangers, Parabolic Troughs, and Phone Casings.
- Developed consumer and industrial products such as Mixture Machines, Helmets, Toothbrushes, Water Taps, and Waste Bins.
- Ensured manufacturability, dimensional accuracy, and functional performance in all designs.
- Collaborated with clients to convert ideas and concepts into practical, cost-effective design solutions.
- **Skills:** 3D Modeling, 3D Rendering, 2D Technical Drawing, Industrial & Product Design Consultation, Clear Communication, Detail Oriented

EDUCATION

Bachelor of Science | University of Engineering and Technology, Peshawar | Khyber Pakhtunkhwa, Pakistan

September 2019 – October 2023

- **Field:** Mechanical Engineering
- **Supervisor:** Professor Usman
- **Final Project:** Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

Faculty of Science | WENSAM college, Dera Ismail Khan | Khyber Pakhtunkhwa, Pakistan

September 2017 – June 2018

- **Field:** Pre-Engineering

PROJECTS

Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

January 2023 – March 2023

- It is our Final Year Project and I have designed its 3D model.
- Automated the manual process of lifting the water dispenser bottle, reducing human efforts.
- Building a customized solution to the problem, starting from lifting to adjustment on water dispenser, atomization
- Use of many mechanisms to optimize results.

3D model of ToothBrush using Solidworks

September 2021 – November 2021

- Modeling of parts: Create a detailed 3D design of the toothbrush
- Assembled: Assemble all component connections accurately
- **Problem:** Traditional toothbrushes contribute to waste and lack proper hygiene.
- **Solution:** CLI Toothbrush features a sustainable design with UV technology that eliminates 99.88% of bacteria and includes a travel-friendly case.
- **Benefits:** Reduces waste, ensures hygiene, and offers a compact, dual-purpose carry case for travel.
- **Functions:** Cleans teeth effectively, sanitizes with UV light, and provides portable protection.

3D model of Mixture Machine using Solidworks

September 2021 – November 2021

- Modeling of parts: Create a detailed 3D design of the mixture machine.
- Assembled: Assemble all component connections accurately
- **Problem:** Bulky, corded blenders; difficult cleaning; not ideal for single-serve use
- **Solution:** Compact, cordless, modular mixer with quick-assembly design
- **Benefits:** Portable, hygienic, space-saving, travel-friendly
- **Functions:** High-speed motor, stainless-steel blade, borosilicate glass jar, safety lock
- **Power:** 2000 mAh USB-C rechargeable battery

3D model of Mini Projects using Solidworks

September 2021 – November 2023

- Modeling of parts: Create a detailed 3D design and 3D rendering images
- Assembled: Assemble all component connections accurately
- **Problem:** Show versatile CAD skills across industrial, consumer, and conceptual designs.
- **Solution:** SolidWorks projects: heat exchanger, pump, solar trough, bottle cap, phone case, sculptural table, Mobius ring.
- **Benefits:** Versatile, manufacturing-aware, parametric, spanning industrial → consumer → abstract forms.
- **Functions:** Fluid power, thermal transfer, solar capture, sealing, device protection, structural support, kinetic demo.

LANGUAGES

- English (Professional)
- Seraiki (Native)
- Urdu & Hindi (Fluent)

CERTIFICATIONS

- Pakistan Engineering Council (PEC), Pakistan
- Certified SOLIDWORKS CAD Design Associate
- Lean Six Sigma White Belt and White Belt (LSSWB and LSSYB)
- Lean Fundamentals and Production
- Data Analytics in Lean Six Sigma
- CPR, AED and First Aid

REFERENCES

- **Professor Dr. Naveed Ahmad**
 - o Assistant Professor, UET, Peshawar, Khyber Pakhtunkhwa, Pakistan
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